TECHNICAL NOTE 3

BLIND SPOTS, ALTERNATIVE DATA & EARLY WARNING + COMPLIANCE PRINCIPALS

Cognitive Biases, OSINT, and Compliance

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1. Blind Spot Analysis: Organisational Cognitive Biases

Zahra and Chaples (1993) identified systematic organisational blind spots that compromise recognised analysis. These cognitive biases can cause organisations to miss critical signals, often with severe consequences. Understanding these biases is the first step toward building a robust intelligence function.

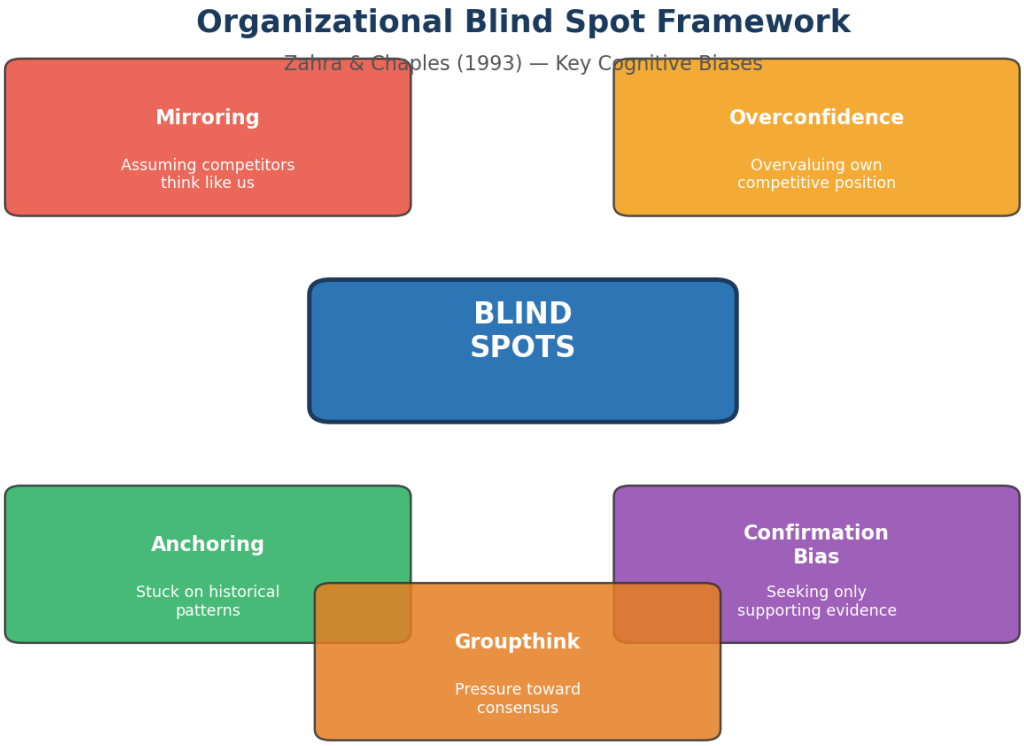

Figure 1. Organisational Blind Spot Framework (adapted from Zahra & Chaples, 1993)

1.1 Types of Blind Spots

Misjudging Industry Boundaries

Organisations often define their competitive arena too narrowly, focusing on traditional competitors while missing threats from adjacent industries, new entrants with different business models, platform players entering multiple industries, and substitute products and services.

EXAMPLE: Blockbuster vs Netflix — The Classic Industry Boundary Blind Spot

In the early 2000s, Blockbuster dominated video rental with over 9,000 stores worldwide. Their competitive analysis focused exclusively on other physical rental chains (Hollywood Video, Movie Gallery). When Netflix launched its DVD-by-mail service in 1997 and streaming in 2007, Blockbuster's leadership dismissed it as a niche offering.

The blind spot: Blockbuster defined their industry as 'video rental' rather than 'home entertainment.' By 2004, when they recognised Netflix as a competitor, they were already losing 3% of their market share annually. Blockbuster filed for bankruptcy in 2010.

CI lesson: *Always define industry boundaries from the customer's perspective ('What job is the customer hiring us to do?'), not from the company's operational perspective.*

Poor Identification of Competition

Common errors include focusing on current competitors while ignoring potential entrants, underestimating small agile competitors, missing non-traditional competitors (customers or suppliers entering your space), and ignoring international competitors not yet in your market.

EXAMPLE: Amazon AWS — When Your Customer Becomes Your Competitor

In 2006, Amazon launched Amazon Web Services (AWS), initially perceived as infrastructure for small startups. Traditional IT companies like IBM, HP, and Oracle focused their CI on each other, ignoring a retailer entering the enterprise technology space.

By 2024, AWS generated over $90 billion in annual revenue, **commanding approximately 31% of the global cloud infrastructure market.** IBM, once the dominant force in enterprise computing, saw its revenue decline from $106 billion (2011) to $62 billion (2023).

CI lesson: *Monitor capabilities, not just current market participation. Ask: 'Who has the resources AND motivation to enter our space?'*

Overemphasis on Visible Competitors

Large, established players receive disproportionate attention while smaller, agile competitors are overlooked. The danger: disruptive innovation often comes from unexpected sources. Nokia's focus on Motorola and Sony Ericsson closed its eyes to Apple's iPhone entry. Kodak watched Fujifilm while digital cameras emerged from consumer electronics companies.

Overconfidence in Historical Patterns

Past competitive behaviours may not predict future actions, especially when new leadership changes competitor culture, technology shifts enable new strategies, market conditions change competitive incentives, or external pressures (investors, regulators) force strategic pivots.

EXAMPLE: Microsoft Under Satya Nadella — When Leadership Changes Everything

Before 2014, competitors could reliably predict Microsoft's behaviour: protect Windows at all costs, resist open-source, and bundle services to maintain the desktop monopoly. When Satya Nadella became CEO, he fundamentally shifted strategy toward cloud-first and mobile-first, embracing Linux, open-sourcing .NET, and acquiring LinkedIn and GitHub.

CI lesson: *Track leadership changes and the backgrounds of new executives. A new CEO from a different industry or function often signals a strategic pivot. Update competitor profiles within 90 days of a C-suite change.*

1.2 Weak Signal Filters

Ilmola and Kuusi (2006) identified filters that block weak signals from reaching decision-makers. These filters operate at structural, cognitive, and cultural levels, creating multiple barriers between market signals and strategic decisions.

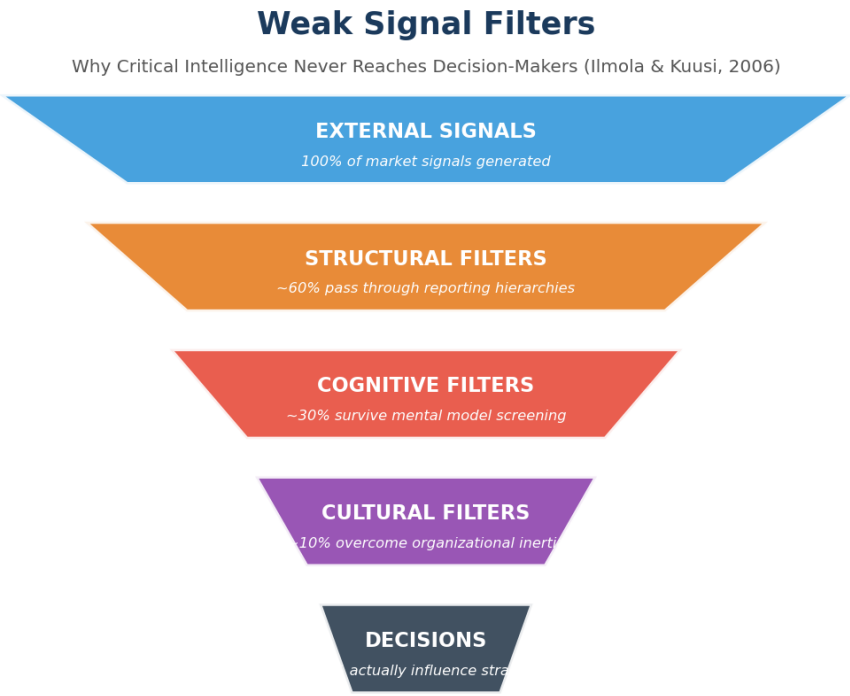

Figure 2. Weak Signal Filter Funnel — from market reality to strategic decisions (Ilmola & Kuusi, 2006)

Structural Filters

- Information systems designed for operational data miss strategic signals
- Reporting hierarchies filter ‘irrelevant’ information before it reaches executives
- Meeting agendas focus on current issues, not emerging threats
- Performance metrics measure what happened, not what is changing

Cognitive Filters

- Mental models cause dismissal of contradictory information
- Confirmation bias leads to seeking supporting evidence
- Anchoring on initial assessments despite new information
- Availability bias overweights recent, dramatic events

Cultural Filters

- 'Not invented here' syndrome rejects external insights
- Success breeds complacency ('we know our industry')
- Hierarchy discourages bad news from rising
- Groupthink suppresses dissenting views

EXAMPLE: Kodak — The Signal That Was Seen But Filtered Out

In 1975, Kodak engineer Steven Sasson invented the first digital camera. Management's response: 'That is cute, but do not tell anyone about it.' Kodak's own engineers generated the weak signal that would destroy their business, but cultural and structural filters prevented it from influencing strategy for nearly 30 years.

All three filter types were active: Structural (R&D budgets allocated to film), Cognitive (executives anchored on chemical imaging model), and Cultural ('film is our DNA' identity blocked adaptation).

1.3 Countermeasures

Organisations can deploy several structured techniques to combat cognitive biases in competitive analysis. The following chart compares their effectiveness against implementation complexity.

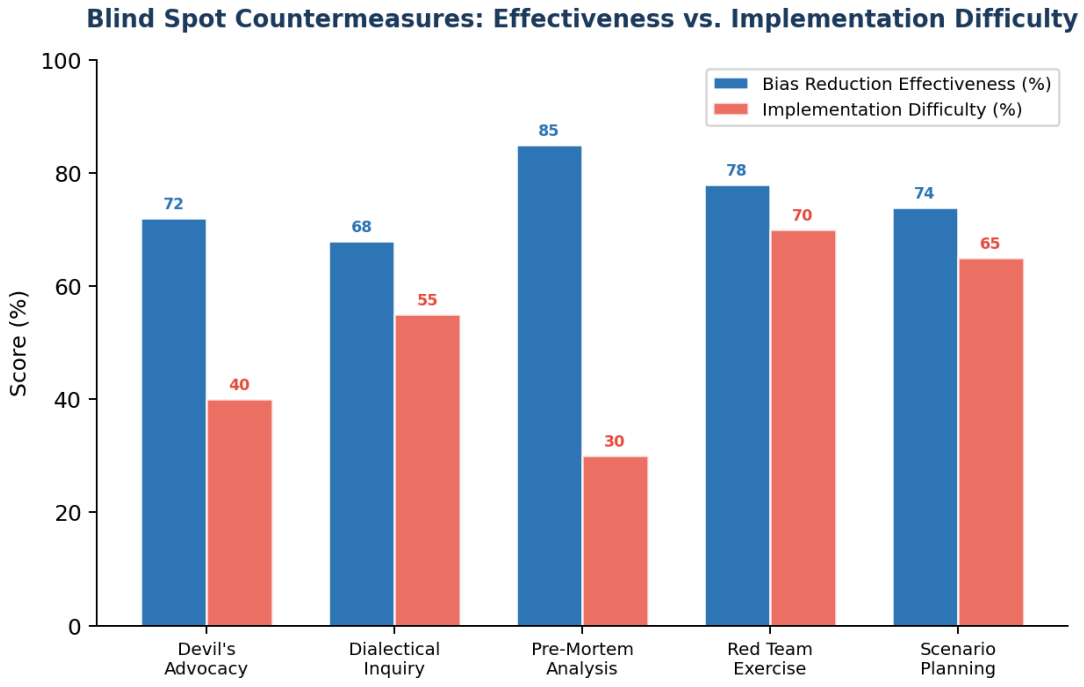

Figure 3. Blind Spot Countermeasures — Effectiveness vs Implementation Difficulty (indicative scores)

Devil's Advocacy

Assign individuals to critique and challenge strategic proposals systematically. The devil's advocate must be empowered to challenge without career risk, prepare a formal critique

document, present the critique before the final decision, and have findings formally addressed in decision documentation.

Dialectical Inquiry

Develop two opposing strategic recommendations based on different underlying assumptions. Team A develops a recommendation based on one set of assumptions; Team B develops a recommendation based on opposing assumptions. A structured debate examines evidence for each position, and decision-makers synthesise insights from both perspectives.

Pre-Mortem Analysis (Klein, 2007)

Imagine the strategy has failed. Work backwards to identify why. Assume it is one year in the future, and the strategy failed spectacularly. Each participant writes down reasons for the failure. Aggregate and discuss failure modes. Identify mitigations for the most likely failure modes.

APPLIED EXAMPLE: Pre-Mortem for a Market Entry Decision

Scenario: A European fintech is considering entering the Southeast Asian mobile payments market.

Pre-mortem results (top failure modes identified):

1. Underestimated local incumbents (GrabPay, GoPay) with deep merchant networks
2. Regulatory complexity: different licensing requirements in each ASEAN country
3. Cultural mismatch: European UX assumptions did not match local user expectations
4. Cash-heavy economies required an offline onramp strategy we had not planned for

Action taken: *Revised the partnership model to use local payment rails rather than direct-to-consumer.*

2. Alternative Data: The New Frontier

Alternative data provides competitive signals unavailable through traditional sources, often revealing competitor activities before official announcements. Originally popularised by hedge funds seeking informational advantages, these data sources are now essential for corporate intelligence functions.

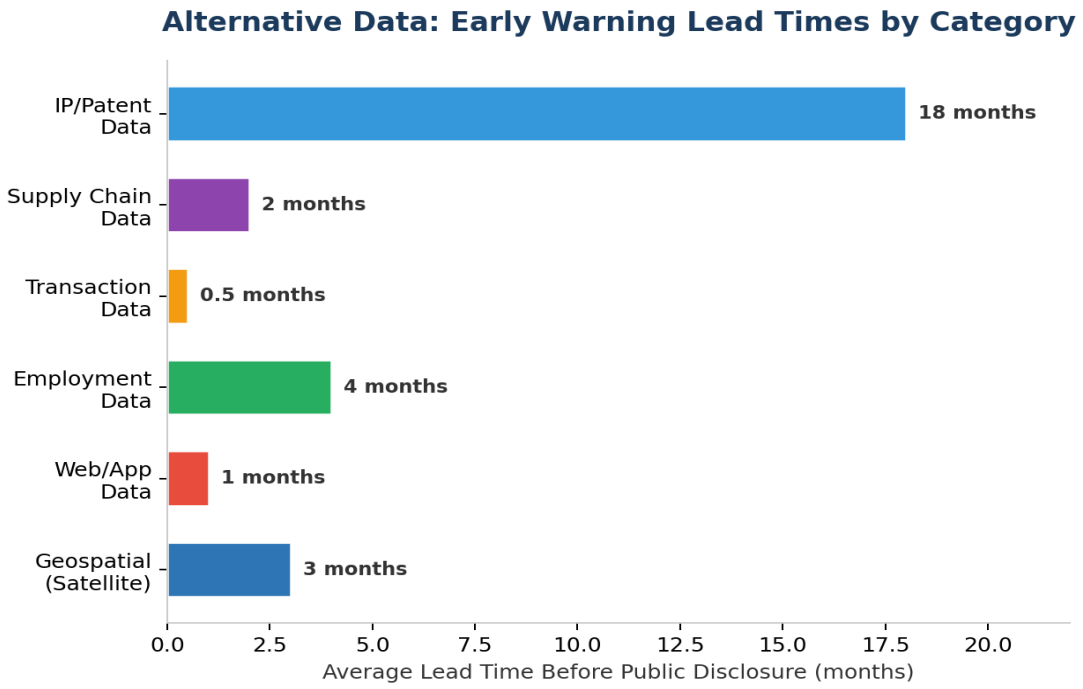

Figure 4. Alternative Data Lead Times — average advance notice before public disclosure by data category

2.1 Satellite Imagery

Providers like Orbital Insight and Planet Labs enable tracking of factory activity (parking lot occupancy, truck traffic, thermal signatures indicate production levels), retail performance (store parking lots predict quarterly revenues with 95%+ accuracy), construction progress, supply chain activity, and agricultural output.

EXAMPLE: Satellite Data Predicts Walmart Revenue

Orbital Insight analysed satellite imagery of parking lots across 500+ Walmart locations, counting cars at regular intervals throughout Q4 2016. Their model predicted Walmart's quarterly revenue within 0.85% accuracy, weeks before the official earnings announcement. Hedge funds using this data could make informed investment decisions before Wall Street analysts published their estimates.

For CI practitioners: *The same technique applies to monitoring competitor factory expansion, retail store performance, and warehouse utilisation in your industry.*

2.2 Web Scraping Intelligence

- **Pricing Changes:** Track competitor pricing across thousands of SKUs daily
- **Product Updates:** Detect new features, discontinued products, and positioning changes
- **Job Postings:** Skill requirements signal strategic direction (e.g., hiring AI engineers = AI product development)
- **anonymisedentiment: Customer reviews reveal product quality issues and competitive positioning**
- **• Website Changes: Messaging shvs**s, new market targeting, leadership announcements

EXAMPLE: Job Postings Reveal Apple's Car Project

In 2015, analysts at recruiting intelligence firms noticed Apple posting hundreds of jobs requiring expertise in automotive engineering, battery technology, and autonomous systems. Many hires came from Tesla, Ford, and Mercedes. This alternative data revealed 'Project Titan'—Apple's secretive automotive initiative—long before any official announcement. The job postings provided a 12-18-month lead time on what would become one of the most discussed projects in tech.

2.3 App Store Intelligence

Mobile app data from Sensor Tower, Apptopia, and similar providers reveals downloads (user acquisition velocity indicating product-market fit), ratings and reviews (customer satisfaction and feature requests), feature updates (product roadmap visibility through release notes), revenue estimates (in-app purchase patterns), and DAU/MAU ratios (engagement metrics revealing product stickiness).

2.4 Credit Card and Transaction Data

Aggregated, anonymised transaction data reveals market share shifts in real time, trends in basket size and average transaction value, shifts between online and offline channels, and competitive switching patterns between brands. GDPR and privacy compliance are essential when using transaction data.

2.5 Location Data

Location intelligence reveals foot traffic (store visits that predict retail performance), dwell time (time spent in-store indicating engagement), cross-shopping (which competitors your customers also visit), and trade area analysis (where customers come from and where they go next).

2.6 Alternative Data Source Comparison

Data Source	Key Signal	Lead Time	Cost Range	Accuracy	Legal Risk
Satellite Imagery	Production & retail activity	1–3 months	$$$	High	Low

Job Postings	Strategic direction	3–6 months	$	Medium	Low
Web Scraping	Pricing & positioning	Days–weeks	$$	High	Medium
App Store Data	Product-market fit	Weeks	$$	Medium	Low
Transaction Data	Revenue & market share	Real-time	$$$$	Very High	High
Patent Filings	R&D direction	18–24 months	$	High	Low

Table 1. Alternative Data Source Comparison Matrix

3. Strategic Early Warning Systems

Effective early warning requires a systematic design of Key Intelligence Topics (KITs) and monitoring indicators. Ben Gilad's Strategic Early Warning (SEW) framework provides the foundation for building intelligence systems that detect weak signals before they become strategic surprises.

3.1 Key Intelligence Topics (KITs)

KITs define what the organization needs to know to make strategic decisions. They structure the intelligence collection process around five dimensions:

1. **Strategic Decisions:** What decisions require external intelligence?
2. **Information Needs:** What information would inform these decisions?
3. **Sources:** Where can this information be obtained?
4. **Timing:** When is the information needed?
5. **Format:** How should intelligence be delivered?

APPLIED EXAMPLE: KIT Design for a Pharmaceutical Company

KIT: Will Competitor X launch a biosimilar version of our top-selling oncology drug?

Decision: Whether to accelerate next-generation formulation or defend the current patent

Sources: FDA ANDA filings, patent challenge (PTAB), job postings for bioprocessing specialists, scientific conference presentations, supplier orders for bioreactor equipment

Timing: Monthly monitoring; escalate immediately if FDA filing detected

3.2 Indicator Types

Leading Indicators (Predict Future Events)

- Patent filings signal R&D direction 18–24 months before product launch
- Executive hiring indicates new strategic initiatives
- Supplier negotiations reveal production plans
- Capital expenditure announcements signal capacity expansion
- Regulatory filings preview market entry

Lagging Indicators (Confirm Past Events)

- Market share confirms competitive positioning
- Revenue and profitability confirm strategy effectiveness
- Customer satisfaction confirms product quality

Lagging indicators are useful for validation but provide no warning.

3.3 Alert System Design

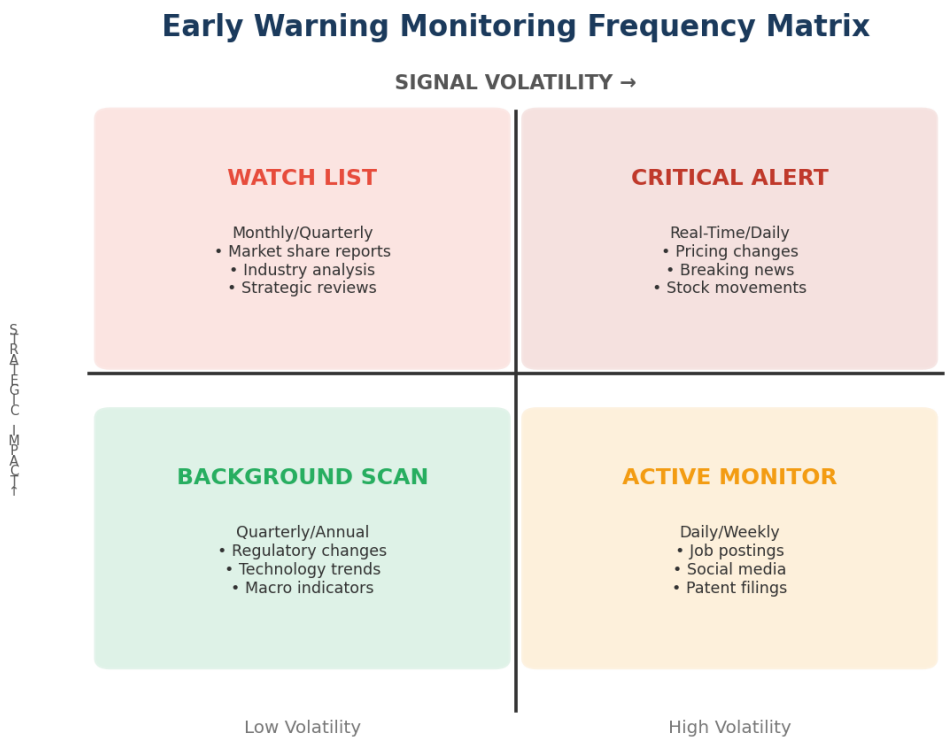

Figure 5. Monitoring Frequency Matrix — match monitoring intensity to signal volatility and strategic impact

- **Alert Thresholds:** Define what level of change triggers action (e.g., competitor price change >5%, unusual patent filing volume)
- **Escalation Protocols:** Who receives alerts? What decisions are they empowered to make?
- **Response Playbooks:** Pre-defined response options for common scenarios
- **False Positive Management:** Process for evaluating and dismissing non-threats

3.4 Monitoring Frequency

Match monitoring frequency to signal volatility:

- **Real-time:** Pricing, stock movements, breaking news
- **Daily:** Social media, job postings, website changes
- **Weekly:** Patent filings, regulatory submissions, app store metrics
- **Monthly:** Market share, industry reports, financial filings
- **Quarterly:** Strategic reviews, competitive landscape reassessment

4. Compliance Framework: Data Collection

CI data collection must operate within legal and ethical boundaries. The consequences of violation include legal liability, reputational damage, and loss of stakeholder trust. This section provides the frameworks necessary for responsible intelligence gathering.

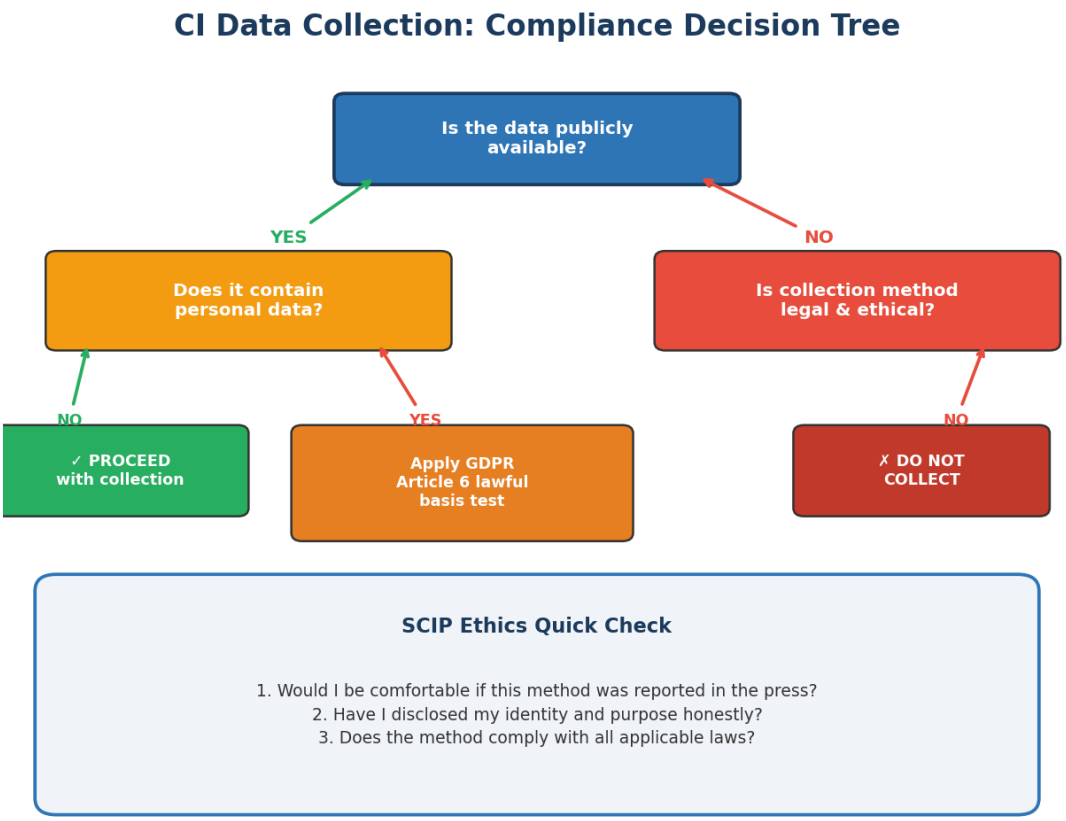

Figure 6. CI Data Collection Compliance Decision Tree

4.1 GDPR Principles for CI

Article 5: Principles Relating to Processing

- **Lawfulness, Fairness, Transparency:** Data must be collected legally with a clear purpose
- **Purpose Limitation:** Data collected for specified purposes cannot be processed incompatibly
- **Data Minimisation:** Collect only what is necessary for the specific purpose
- **Accuracy:** Take reasonable steps to ensure data is accurate and up to date
- **Storage Limitation:** Keep data only as long as necessary
- **Integrity and Confidentiality:** Ensure appropriate security of personal data

Lawful Bases for Processing (Article 6)

- Consent: Individual has given clear consent
- Contract: Processing necessary for contract performance
- Legal Obligation: Processing required by law
- Vital Interests: Necessary to protect someone's life
- Public Task: Processing necessary for public interest
- Legitimate Interests: Necessary for legitimate interests unless overridden by data subject rights

4.2 SCIP Code of Ethics

The Strategic and Competitive Intelligence Professionals (SCIP) Code of Ethics requires practitioners to:

6. Continually strive to increase the recognition and respect of the profession
7. Comply with all applicable laws, domestic and international
8. Accurately disclose all relevant information, including identity and organisation, prior to all interviews
9. Avoid conflicts of interest in fulfilling duties
10. Provide honest and realistic recommendations and conclusions
11. Promote this code of ethics within the company, with third-party contractors, and within the entire profession

4.3 Ethical Grey Areas

Some practices are legal but ethically questionable. The following table provides guidance:

Practice	Acceptable	Not Acceptable
Trade Show Intelligence	Attending competitor booths openly	Misrepresenting identity or purpose
Job Interviews	Interviewing competitor employees for legitimate positions	'Fake' interviews to extract confidential information
Social Engineering	Gathering information from public sources	Deceiving employees to reveal confidential data
Reverse Engineering	Analysing publicly available products	Violating licensing agreements
Web Scraping	Collecting publicly posted pricing data	Circumventing access controls or scraping personal data

Table 2. Ethical Grey Areas in Competitive Intelligence — Guidance Matrix

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